$A \in M_{n \times n}$

$v \neq 0$ is an eigenvector if $Av = \lambda v$
where $\lambda \in \mathbb{R}$ is its eigenvalue.

Theorem

$\lambda \in \mathbb{R}$ is an eigenvalue of A

$\iff \det(A - \lambda I) = 0$

$\det(A - xI)$ is the characteristic polynomial of A

Remark

- $\det(A - xI) = (-1)^n x^n + \cdots + b_1 x + b_0$ for some $b_0, \ldots, b_{n-1} \in \mathbb{R}$
  a polynomial of degree n and leading term $(-1)^n$.

- Computing $\det(A - xI)$ is tedious. Finding solutions to $\det(A - xI) = 0$ is nearly impossible for $n \geq 5$.

- Take $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Then $\det(A - xI)$

$$= \det \begin{bmatrix} -x & 1 \\ -1 & -x \end{bmatrix}$$

$$= (-x)^2 - (-1)(1)$$

$$= x^2 + 1$$

The characteristic equation $x^2 + 1 = 0$

has no real solutions! Consequently

A has no eigenvalues/eigenvectors.

- If $\lambda \in \mathbb{R}$ is an eigenvalue of A then it is a root of $\det(A - xI)$. By the division algorithm $\det(A - xI) = (x-\lambda)^1 p(x)$ for some $p(x)$ of degree $n-1$. There is a highest power $m \geq 1$ such that

$$\det(A - xI) = (x-\lambda)^m q(x)$$

where $(x-\lambda)$ does not divide $q(x)$ ($q(\lambda) \neq 0$)

Def. The (algebraic) multiplicity of an eigenvalue λ of A is the highest power m of $(x-\lambda)$ occurring as a factor in $\det(A-xI)$

- If $A \in M_{n\times n}$ has exactly k distinct eigenvalues $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ with multiplicities $m_1, \dots, m_k$ then

$$\det(A - xI) = (x-\lambda_1)^{m_1}(x-\lambda_2)^{m_2}\cdots(x-\lambda_k)^{m_k} p(x)$$

where $p(x)$ has degree $n-(m_1+\cdots+m_k)$.

Comparing degrees we see that

$$n \geq m_1 + \cdots + m_k \geq k = \#\text{ of eigenvalues}$$

Example Take $A = \begin{bmatrix} 5 & -2 & 6 \\ 0 & 3 & -8 \\ 0 & 0 & 5 \end{bmatrix}$ with

$$\det(A - xI) = -(x-5)^2(x-3)^1$$

$\Rightarrow$ multiplicity of 5 is 2
multiplicity of 3 is 1

Def Two matrices $A, B \in M_{n\times n}$ are <u>similar</u> if there is an invertible $P \in M_{n\times n}$ such that $A = PBP^{-1}$

Example

Take $B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Then $P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ and

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$$

$\Rightarrow$ $A = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$ is similar to $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Remarks

- Take $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. Then $PBP^{-1} = P0P^{-1}$
$= 0$.

The only matrix similar to the zero matrix is the zero matrix itself.

- Take $B = I$. Then $PIP^{-1} = PP^{-1} = I$.

  The only matrix similar to I is I itself.

- Take $P = I = P^{-1}$. Then $PBP^{-1} = IBI$
  $= B$.

  B is similar to itself!

- observe that $A = PBP^{-1} \iff P^{-1}AP = P^{-1}PBP^{-1}P$
  $= IBI$
  $= B$.

  $\Rightarrow B = P^{-1}AP = (P^{-1})A(P^{-1})^{-1}$

  "A similar to $B \iff B$ similar to A".

Theorem

If A and B are similar matrices then

they $\det(A - xI) = \det(B - xI)$

and so A and B have identical

eigenvalues with identical multiplicities.

Sketch of proof. Suppose $A = PBP^{-1}$.

$\det(A - xI) = \det(PBP^{-1} - xI)$

$$= \det(PBP^{-1} - xPP^{-1})$$
$$= \det(PBP^{-1} - P(xI)P^{-1})$$
$$= \det(P(B - xI)P^{-1})$$
$$= \det(P)\det(B - xI)\det(P^{-1})$$
$$= \det(P)\det(P^{-1})\det(B - xI) \quad \text{(from 36)}$$
$$= \det(PP^{-1})\det(B - xI)$$
$$= \cancel{\det(I)}^{1}\det(B - xI)$$
$$= \det(B - xI)$$

Aside: $\det(P)\det(P^{-1}) = 1$

$$\Rightarrow \det(P^{-1}) = \frac{1}{\det(P)} = (\det(P))^{-1}$$

5.3 Diagonalization

$A \in M_{n \times n}$.

Def. $A \in M_{n \times n}$ is <u>diagonalizable</u> if

A is similar to a diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$ (off diagonal entries are zero).

i.e. $A = PDP^{-1} = P \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} P^{-1}$.

Theorem 5 The following are equivalent.

(i) $A \in M_{n\times n}$ is diagonalizable.

(ii) A has n linearly independent eigenvectors $v_1, \dots, v_n \in \mathbb{R}^n$.

(iii) $\mathbb{R}^n$ has a basis $\{v_1, \dots, v_n\}$ of eigenvectors of A.

(iv) $A = PDP^{-1}$ where $P = [v_1 \cdots v_n]$

$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$, and $\{v_1, \dots, v_n\}$ is a basis of eigenvectors with $Av_j = \lambda_j v_j$ $1 \le j \le n$.

Sketch of proof.

(ii) is equivalent to (iii) since $\dim \mathbb{R}^n = n$ and (Thm 13 §4.5)

"(i) $\Rightarrow$ (iv)" Suppose A is diagonalizable as the defn $A = PDP^{-1}$ where $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$

Notice that $A = PDP^{-1} \iff AP = PD$.

Let $P = [v_1 \cdots v_n]$, i.e. v_j is the jth column of P.

Then $AP = A[v_1 \cdots v_n] = [Av_1 \cdots Av_n]$

Moreover $PD = P\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} = \left[P\begin{bmatrix} \lambda_1 \\ \vdots \\ 0 \end{bmatrix} \cdots P\begin{bmatrix} 0 \\ \vdots \\ \lambda_n \end{bmatrix} \right]$

$= [\lambda_1 v_1 \cdots \lambda_n v_n]$

$$\Rightarrow \quad [Av_1 \cdots Av_n] = [\lambda_1 v_1 \ \cdots \ \lambda_n v_n]$$

$$\Rightarrow \quad Av_1 = \lambda_1 v_1, \ Av_2 = \lambda_2 v_2, \ \ldots, \ Av_n = \lambda_n v.$$

By Thm §2.3, since P is invertible its columns $\{v_1, \ldots, v_n\}$ form a basis of $\mathbb{R}^n$. In particular $v_j \neq 0$ for any $(1 \leq) j \leq n$.

This concludes the proof of (i) $\Rightarrow$ (iv)

"(iii) $\Rightarrow$ (iv)" By (iii) $Av_1 = \lambda_1 v_1, \ldots, Av_n = \lambda_n v_n$

write $P = [v_1 \cdots v_n]$, $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$

and show $AP = PD$ as above.